

Alberto Archetti, Ph.D.

POSTDOCTORAL RESEARCHER IN ARTIFICIAL INTELLIGENCE



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Biography

I am fascinated by emerging phenomena — how simple rules give rise to complex behavior, whether in vitro or in silico — and I find the field of artificial intelligence the ideal playground to study them.

As a postdoctoral researcher at AIRLab, Politecnico di Milano, I work on **tabular foundation models**, **federated learning**, and **survival analysis**, building models that learn from sensitive, multimodal data — turning scattered information into reliable, trustworthy predictions. I also love teaching and mentoring, and I take part in educational initiatives to promote AI literacy at all levels.

Research Interests

- **Tabular Foundation Models** — Building general-purpose deep learning models for in-context learning on tabular data.
- **Survival Analysis** — Predicting time-to-event outcomes for clinical data and financial applications.
- **Federated Learning** — Training models across distributed, siloed datasets while preserving privacy.
- **Machine Learning for Healthcare** — Applying AI to multimodal healthcare data, with a focus on privacy, interpretability, and reliability.

Publications

161 citations · h-index 7 · [Google Scholar](#)

Journal Articles

- [1] Archetti, Alberto, Giusti, G., Gorla, K. R., Caputo, S., Magarini, M., Matteucci, M., Mucchi, L., Pierobon, M. (2025). Emergent Molecular Communication: Preliminary Results With Graph Neural Networks and Diffusion Channels. *IEEE Transactions on Molecular, Biological, and Multi-Scale Communications*.
- [2] Archetti, Alberto, Stranieri, F., Matteucci, M. (2024). Bridging the gap: improve neural survival models with interpolation techniques. *Progress in Artificial Intelligence*, pp. 1–16.
- [3] Archetti, Alberto, Ieva, F., Matteucci, M. (2023). Scaling survival analysis in healthcare with federated survival forests: A comparative study on heart failure and breast cancer genomics. *Future Generation Computer Systems*, 149, pp. 343–358. [10.1016/j.future.2023.07.036](https://doi.org/10.1016/j.future.2023.07.036)

Conference Papers

- [1] Erbil, P., Archetti, Alberto, Lomurno, E., Matteucci, M. (2026). Deep Variational Contrastive Learning for Joint Risk Stratification and Time-to-Event Estimation. 2026 International Joint Conference on Neural Networks (IJCNN).
- [2] Mastroleo, M., Archetti, Alberto, Mastroleo, F., Matteucci, M. (2026). SurvKAN: A Fully Parametric Survival Model Based on Kolmogorov-Arnold Networks. 2026 International Joint Conference on Neural Networks (IJCNN).
- [3] Archetti, Alberto, Lomurno, E., Piccinotti, D., Matteucci, M. (2025). FPBoost: Fully Parametric Gradient Boosting for Survival Analysis. *European Conference on Artificial Intelligence (ECAI)*.
- [4] Alberto Archetti (2025). Federated Survival Analysis: Ensemble and Neural Methods for Distributed Time-to-Event Data.
- [5] Menta, A., Archetti, Alberto, Matteucci, M. (2024). Latent neural cellular automata for resource-efficient image restoration. *Artificial Life Conference Proceedings* 36, 2024.

- [6] Archetti, Alberto, Matteucci, M. (2023). Federated Survival Forests. 2023 International Joint Conference on Neural Networks (IJCNN), pp. 1–9. [10.1109/IJCNN54540.2023.10190999](https://doi.org/10.1109/IJCNN54540.2023.10190999)
- [7] Lomurno, E., Archetti, Alberto, Ausonio, F., Matteucci, M. (2023). Discriminative adversarial privacy: balancing accuracy and membership privacy in neural networks. The 34th British Machine Vision Conference Proceedings (BMVC).
- [8] Lomurno, E., Archetti, Alberto, Cazzella, L., Samele, S., Di Perna, L., Matteucci, M. (2023). SGDE: Secure Generative Data Exchange for Cross-Silo Federated Learning. Proceedings of the 2022 5th International Conference on Artificial Intelligence and Pattern Recognition, pp. 205–214. [10.1145/3573942.3573974](https://doi.org/10.1145/3573942.3573974)
- [9] Archetti, Alberto, Cannici, M., Matteucci, M. (2022). Neural Weighted A*: Learning Graph Costs and Heuristics with Differentiable Anytime A*. Machine Learning, Optimization, and Data Science (LOD 2021), pp. 596–610. [10.1007/978-3-030-95467-3_43](https://doi.org/10.1007/978-3-030-95467-3_43)

Workshop Papers

- [1] Archetti, Alberto, Lomurno, E., Lattari, F., Martin, A., Matteucci, M. (2023). Heterogeneous Datasets for Federated Survival Analysis Simulation. Companion of the 2023 ACM/SPEC International Conference on Performance Engineering, pp. 173–180. [10.1145/3578245.3584935](https://doi.org/10.1145/3578245.3584935)
- [2] Archetti, Alberto, Stranieri, F., Matteucci, M. (2023). Deep Survival Analysis for Healthcare: An Empirical Study on Post-Processing Techniques. CEUR Workshop Proceedings, 3578, pp. 99–121.
- [3] Stranieri, F., Archetti, Alberto, Robbiano, E., Kouki, C., Stella, F. (2023). Drug Inventory Control: Human Decisions versus Deep Reinforcement Learning. CEUR Workshop Proceedings.

Preprints

- [1] Sabella, M., Archetti, Alberto, Pinoli, P., Matteucci, M., Cappiello, C. (2026). QuAIL: Quality-Aware Inertial Learning for Robust Training under Data Corruption. [2602.03686](https://arxiv.org/abs/2602.03686)

Experience

2025–present	Research Fellow AIRLab, Politecnico di Milano · Milan, Italy RESEARCH TOPIC — Tabular foundation models for healthcare.
2021–2025	National Ph.D. in Artificial Intelligence for Industry 4.0 Politecnico di Torino · Turin, Italy RESEARCH TOPIC — Federated Learning for Survival Analysis. COLLABORATIONS — Research visitor at the Human Technopole, Milan, Italy.
2021	Research Internship Politecnico di Milano · Milan, Italy ACTIVITIES — Data analysis for anomaly detection; automated knowledge graph extraction.

Projects

2021–2023	AI-SPRINT — Artificial Intelligence in Secure PRivacy-preserving computing coNTinuum European Commission · Horizon 2020 (GA 101016577) · Politecnico di Milano, Milan, Italy ROLE — Federated Learning advisor
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Education

2021–2025	National Ph.D. in Artificial Intelligence for Industry 4.0 Politecnico di Torino · Italy THESIS — Federated Survival Analysis: Ensemble and Neural Methods for Distributed Time-to-Event Data
2018–2021	Master of Science in Computer Science and Engineering Politecnico di Milano · Italy · 110/110 cum Laude THESIS — Neural Weighted A*: Learning Graph Costs and Heuristics with Differentiable Anytime A*
2015–2018	Bachelor's Degree in Computer Science and Engineering Politecnico di Milano · Italy · 110/110 cum Laude
2010–2015	Diploma di Liceo Scientifico Madonna della Neve · Italy · 100/100

Skills

- **Programming** — Python, C/C++, Java, SQL, LaTeX, Bash
- **Machine Learning** — PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, n8n
- **Tools** — Git, Docker, Linux, SLURM/HPC
- **Languages** — Italian (native), English (fluent)

Teaching Activities

2026	Advanced Deep Learning (20h) Teaching Assistant · Politecnico di Milano · MSc
2025	AI Bootcamp (15h) Teaching Assistant · Politecnico di Milano · TechCamp@PoliMI
2025	AI Product Management Bootcamp (6h) Teaching Assistant · CEFRIEL · Corporate Training
2025	Artificial Neural Networks and Deep Learning (20h) Teaching Assistant · Politecnico di Milano · MSc
2024	Artificial Neural Networks and Deep Learning (20h) Teaching Assistant · Politecnico di Milano · MSc
2024	AI Bootcamp (15h) Teaching Assistant · Politecnico di Milano · TechCamp@PoliMI
2024	Coding Bootcamp (15h) Teaching Assistant · Politecnico di Milano · TechCamp@PoliMI
2024	Software Engineering (32h) Lab Tutor · Politecnico di Milano · BSc
2023	Coding Bootcamp (15h) Teaching Assistant · Politecnico di Milano · TechCamp@PoliMI
2023	Software Engineering (32h) Lab Tutor · Politecnico di Milano · BSc
2022	Coding Bootcamp (30h) Teaching Assistant · Politecnico di Milano · TechCamp@PoliMI
2022	Software Engineering (32h) Lab Tutor · Politecnico di Milano · BSc

Supervision

2026	Pinar Erbil MSc thesis · Co-advisor Deep Variational Contrastive Learning for Risk Stratification and Time-to-Event Estimation
2026	Marina Mastroleo MSc thesis · Co-advisor SurvKAN: Fully Parametric Survival Modeling with Kolmogorov-Arnold Networks
2026	Sofia Perini MSc thesis · Co-advisor Exploring multimodality in federated survival analysis
2025	Emanuele Paesano MSc thesis · Co-advisor A multimodal framework for survival analysis integrating clinical, genomic, histopathological and textual data
2025	Niccolò Maria Rizzi MSc thesis · Co-advisor A generative pipeline for high-quality synthetic survival datasets
2025	Gabriele Giusti MSc thesis · Co-advisor Multi-Agent Reinforcement Learning for emergent molecular communication in diffusion-based environments
2023	Andrea Menta MSc thesis · Co-advisor Image restoration via Latent Neural Cellular Automata
2022	Simone Cimmino MSc thesis · Co-advisor A pipeline for company industrial sector classification from unstructured website content
2022	Sara Sacco MSc thesis · Co-advisor Analisi spettrale per il rilevamento di anomalie: studio sulla sopravvivenza degli estensimetri a corda per Snam S.p.A.

Talks

2025	FPBoost: Fully Parametric Gradient Boosting for Survival Analysis European Conference on Artificial Intelligence (ECAI) · Bologna, Italy · Conference talk
2024	Discriminative adversarial privacy: balancing accuracy and membership privacy in neural networks British Machine Vision Conference (BMVC) · Aberdeen, Scotland · Conference talk
2023	Federated Survival Forests International Joint Conference on Neural Networks (IJCNN) · Broadbeach, Australia · Conference talk
2023	Deep Survival Analysis for Healthcare: An Empirical Study on Post-Processing Techniques AlxIA Workshop on Artificial Intelligence For Healthcare · Rome, Italy · Conference talk
2022	Federated Survival Analysis AlxIA Workshop on Machine Learning and Data Mining · Udine, Italy · Conference talk
2022	Federated and Privacy-Preserving Learning PhD Lecture, Politecnico di Milano · Milan, Italy · Seminar
2021	Neural Weighted A*: Learning Graph Costs and Heuristics with Differentiable Anytime A* Conference on Machine Learning, Optimization, and Data Science (LOD) · Grasmere, England · Conference talk

Awards and Achievements

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| 2021 | Special Mention for Best Paper
LOD 2021 – 7th International Conference on Machine Learning, Optimization, and Data Science |
| 2021 | Honours Programme: Scientific Research in Information Technology
Politecnico di Milano |

Data Authorization

Autorizzo al trattamento dati ai sensi del GDPR 2016/679 del 27 aprile 2016 (Regolamento Europeo relativo alla protezione delle persone fisiche per quanto riguarda il trattamento dei dati personali). Autorizzo la pubblicazione del Curriculum Vitae sul sito istituzionale del Politecnico di Milano (sez. Amministrazione Trasparente) in ottemperanza al D. Lgs n. 33 del 14 marzo 2013 (e s.m.i.).